



3 Complex Analysis 2024

Tutorial 6

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Consequences of uniform convergence

Problem 24 Let $G \subseteq \mathbb{C}$ be a region, let γ be a path in G , and let $(f_n)_{n=1}^{\infty}$ be a sequence of continuous functions $f_n : G \rightarrow \mathbb{C}$.

- (a) Suppose that f_n converges to $f : \gamma \rightarrow \mathbb{C}$ uniformly on γ . We know that f is continuous on γ . Prove that

$$\lim_{n \rightarrow \infty} \int_{\gamma} f_n(z) dz = \int_{\gamma} f(z) dz.$$

- (b) Suppose that for each $n \in \mathbb{N}$ there is $M_n \geq 0$ such that $|f_n(z)| \leq M_n$ for all $z \in \gamma$, and that the series $\sum_{n=1}^{\infty} M_n$ converges. Prove that

$$\sum_{n=1}^{\infty} \int_{\gamma} f_n(z) dz = \int_{\gamma} \sum_{n=1}^{\infty} f_n(z) dz.$$

Remark. This statement is used in the proof of Taylor's Theorem.

Solution.

- (a) By the uniform convergence, given $\varepsilon > 0$ there is $N \in \mathbb{N}$ such that

$$|f_n(z) - f(z)| < \varepsilon \quad \text{for all } z \in \gamma, n \geq N.$$

But then

$$\left| \int_{\gamma} f_n(z) dz - \int_{\gamma} f(z) dz \right| = \left| \int_{\gamma} (f_n(z) - f(z)) dz \right| \leq \varepsilon \ell(\gamma), \quad n \geq N,$$

and the statement follows.

- (b) Put $f = \sum_{n=1}^{\infty} f_n$. We have to show that

$$\sum_{n=1}^{\infty} \int_{\gamma} f_n(z) dz = \int_{\gamma} f(z) dz.$$

For each partial sum we have

$$\sum_{n=1}^N \int_{\gamma} f_n(z) dz = \int_{\gamma} \left(\sum_{n=1}^N f_n(z) \right) dz.$$

By the Weierstrass M-Test, the sequence of partial sums $\left(\sum_{n=1}^N f_n\right)_{N=1}^{\infty}$ converges uniformly to f . By (a),

$$\sum_{n=1}^{\infty} \int_{\gamma} f_n(z) dz = \lim_{N \rightarrow \infty} \sum_{n=1}^N \int_{\gamma} f_n(z) dz = \lim_{N \rightarrow \infty} \int_{\gamma} \left(\sum_{n=1}^N f_n(z) \right) dz = \int_{\gamma} f(z) dz$$

as desired.

Cauchy's integral formula

Problem 25 Evaluate the integral

$$\frac{1}{2\pi} \int_{\gamma} \frac{e^z}{z^2 + 1} dz,$$

where

- (a) $\gamma(t) = i + e^{it}$, $t \in [0, 2\pi]$,
- (b) $\gamma(t) = -i + e^{it}$, $t \in [0, 2\pi]$,
- (c) $\gamma(t) = 3e^{it}$, $t \in [0, 2\pi]$.

Hint: Partial fraction decomposition might help.

Solution.

Let us first find the partial fraction decomposition of $\frac{1}{z^2+1} = \frac{1}{(z-i)(z+i)}$. We have

$$\frac{1}{(z-i)(z+i)} = \frac{A}{z-i} + \frac{B}{z+i} = \frac{(A+B)z + (A-B)i}{(z-i)(z+i)},$$

so that $A+B=0$, $(A-B)i=1$. The solution is $A=-\frac{i}{2}$, $B=\frac{i}{2}$, i.e.

$$\frac{1}{(z-i)(z+i)} = \frac{i}{2} \left(\frac{1}{z+i} - \frac{1}{z-i} \right).$$

- (a) If $\gamma(t) = i + e^{it}$, $t \in [0, 2\pi]$, then only $z = i$ lies inside the circle. By Cauchy's Integral Formula,

$$\frac{1}{2\pi} \int_{\gamma} \frac{e^z}{z^2 + 1} dz = i \frac{1}{2\pi i} \int_{\gamma} \frac{(e^z/(z+i))}{z-i} dz = i \cdot \frac{e^i}{2i} = \frac{e^i}{2}.$$

- (b) If $\gamma(t) = -i + e^{it}$, $t \in [0, 2\pi]$, then only $z = -i$ lies inside the circle. By Cauchy's Integral Formula,

$$\frac{1}{2\pi} \int_{\gamma} \frac{e^z}{z^2 + 1} dz = i \frac{1}{2\pi i} \int_{\gamma} \frac{(e^z/(z-i))}{z+i} dz = i \cdot \frac{e^{-i}}{-2i} = -\frac{e^{-i}}{2}.$$

(c) If $\gamma(t) = 3e^{it}$, $t \in [0, 2\pi]$, then both points $z = \pm i$ lie inside the circle. We have

$$\begin{aligned} \frac{1}{2\pi} \int_{\gamma} \frac{e^z}{z^2 + 1} dz &= i \frac{1}{2\pi i} \frac{1}{2} \left(\int_{\gamma} \frac{e^z}{z + i} dz - \int_{\gamma} \frac{e^z}{z - i} dz \right) \\ &= -\frac{1}{2}(e^{-i} - e^i) = -i \frac{e^i - e^{-i}}{2i} = -i \sin 1. \end{aligned}$$

Problem 26 Let $\gamma(t) = 2e^{it}$, $t \in [0, 2\pi]$. Evaluate the following integrals.

$$(a) \int_{\gamma} \frac{z^3 + 5}{z - i} dz, \quad (b) \int_{\gamma} \frac{1}{z^2 + z + 1} dz, \quad (c) \int_{\gamma} \frac{\sin z}{z^2 - 1} dz.$$

Solution.

(a) We have

$$\int_{\gamma} \frac{z^3 + 5}{z - i} dz = 2\pi i(i^3 + 5) = 2\pi i(5 - i).$$

(b) We have $z^2 + z + 1 = (z - z_1)(z - z_2)$, where $z_1 = -\frac{1}{2} - i\frac{\sqrt{3}}{2}$, $z_2 = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$. Both points z_1, z_2 lie inside the circle γ . We use the partial fraction decomposition

$$\frac{1}{z^2 + z + 1} = \frac{i}{\sqrt{3}} \left(\frac{1}{z - z_1} - \frac{1}{z - z_2} \right).$$

We have

$$\int_{\gamma} \frac{1}{z^2 + z + 1} dz = \frac{i}{\sqrt{3}} \left(\int_{\gamma} \frac{1}{z - z_1} dz - \int_{\gamma} \frac{1}{z - z_2} dz \right) = \frac{i}{\sqrt{3}}(2\pi i - 2\pi i) = 0.$$

(c) We have $z^2 - 1 = (z - 1)(z + 1)$ and both points $z = -1$ and $z = 1$ lie inside the circle γ . The partial fraction decomposition is

$$\frac{1}{z^2 - 1} = \frac{1}{2} \left(\frac{1}{z - 1} - \frac{1}{z + 1} \right).$$

Thus,

$$\int_{\gamma} \frac{\sin z}{z^2 - 1} dz = \frac{1}{2} \left(\int_{\gamma} \frac{\sin z}{z - 1} dz - \int_{\gamma} \frac{\sin z}{z + 1} dz \right) = \frac{1}{2} \cdot 2\pi i(\sin 1 - \sin(-1)) = 2\pi i \sin 1.$$

Problem 27 Evaluate the integral

$$\int_{\gamma} \frac{\operatorname{Re} z}{z - \frac{1}{2}} dz,$$

where $\gamma(t) = e^{it}$, $t \in [0, 2\pi]$.

Hint: On the unit circle, the function $\operatorname{Re} z$ coincides with a holomorphic function.

Solution.

We have $\operatorname{Re} z = \frac{1}{2}(z + \bar{z})$. If $|z| = 1$, then $\bar{z} = \frac{1}{z}$ (this follows from $|z|^2 = z\bar{z} = 1$), and thus $\operatorname{Re} z = \frac{1}{2}\left(z + \frac{1}{z}\right)$. Our integral is

$$\int_{\gamma} \frac{\operatorname{Re} z}{z - \frac{1}{2}} dz = \frac{1}{2} \left(\int_{\gamma} \frac{z}{z - \frac{1}{2}} dz + \int_{\gamma} \frac{1}{z\left(z - \frac{1}{2}\right)} dz \right).$$

For the first integral we have

$$\int_{\gamma} \frac{z}{z - \frac{1}{2}} dz = 2\pi i \cdot \frac{1}{2} = \pi i.$$

For the second integral we use the partial fraction decomposition

$$\frac{1}{z\left(z - \frac{1}{2}\right)} = 2 \left(\frac{1}{z - \frac{1}{2}} - \frac{1}{z} \right)$$

and obtain

$$\int_{\gamma} \frac{1}{z\left(z - \frac{1}{2}\right)} dz = 2 \left(\int_{\gamma} \frac{1}{z - \frac{1}{2}} dz - \int_{\gamma} \frac{1}{z} dz \right) = 2(2\pi i - 2\pi i) = 0.$$

Thus,

$$\int_{\gamma} \frac{\operatorname{Re} z}{z - \frac{1}{2}} dz = \frac{1}{2}\pi i.$$